Characterizing Double White Dwarf Star Systems

By G. Leigh Harwell

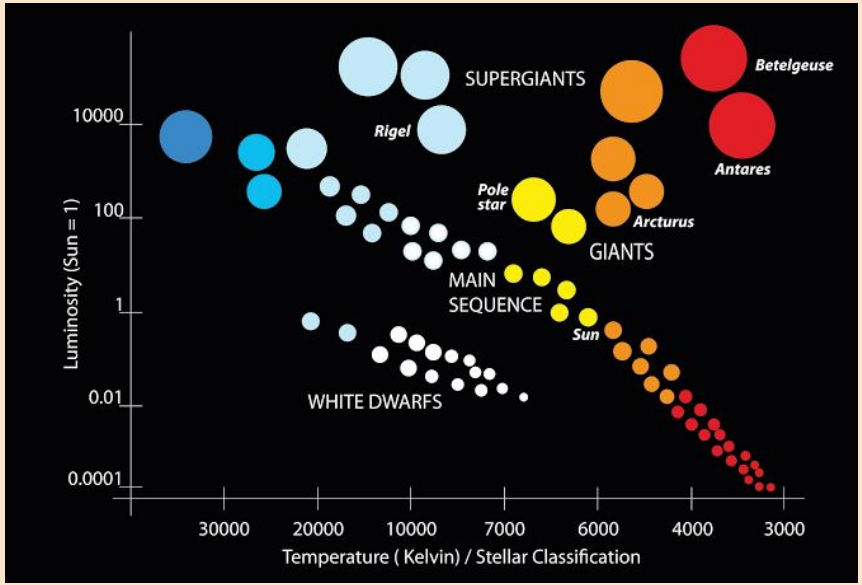


Double White Dwarf System

White Dwarf (WD) stars are the dense leftovers of dying stars that have run out of nuclear fuel

- Found in the low luminosity and high temperature region of the Hertzsprung-Russell (H-R) diagram

Double White Dwarf (DWD) system is a binary system of a seen primary and unseen secondary white dwarf star



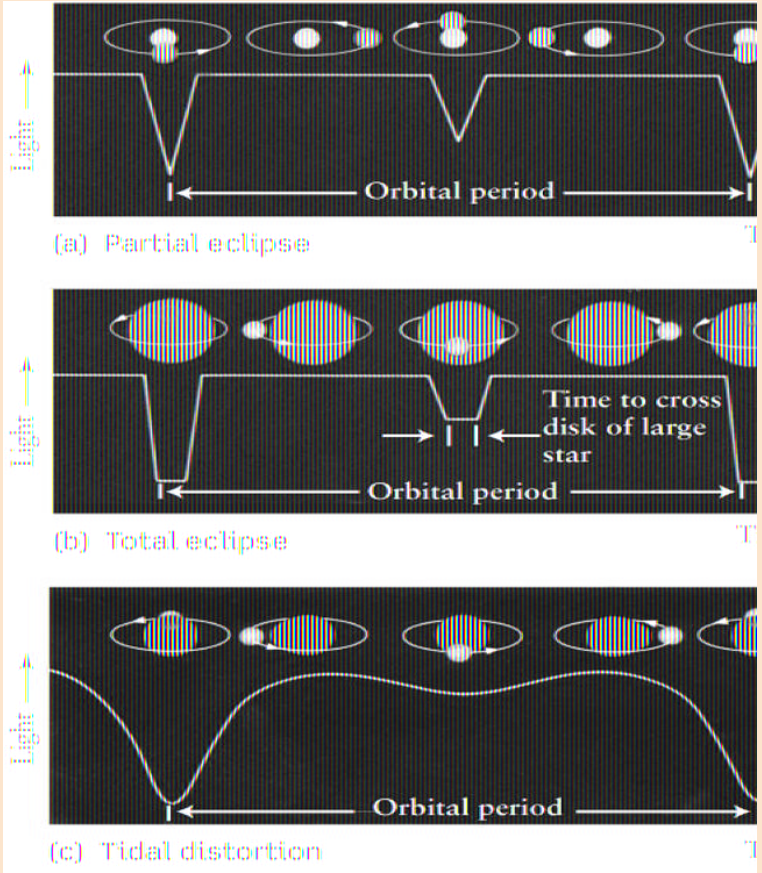
Simple HR Diagram
Temperature on the y-axis with the right side being higher temperatures and Luminosity on the x-axis

Kupfer, Thomas, et al. "LISA Galactic Binaries with Astrometry from Gaia DR3." arXiv preprint, arXiv:2302.12719, 2023

Double White Dwarf System

DWD system is a binary system of a seen primary and unseen secondary white dwarf star

- Some of the most abundant compact binaries in the Milky Way
- Good for studying binary evolution
- These systems with short orbital periods are predicted to be strong gravitational wave (GW) source



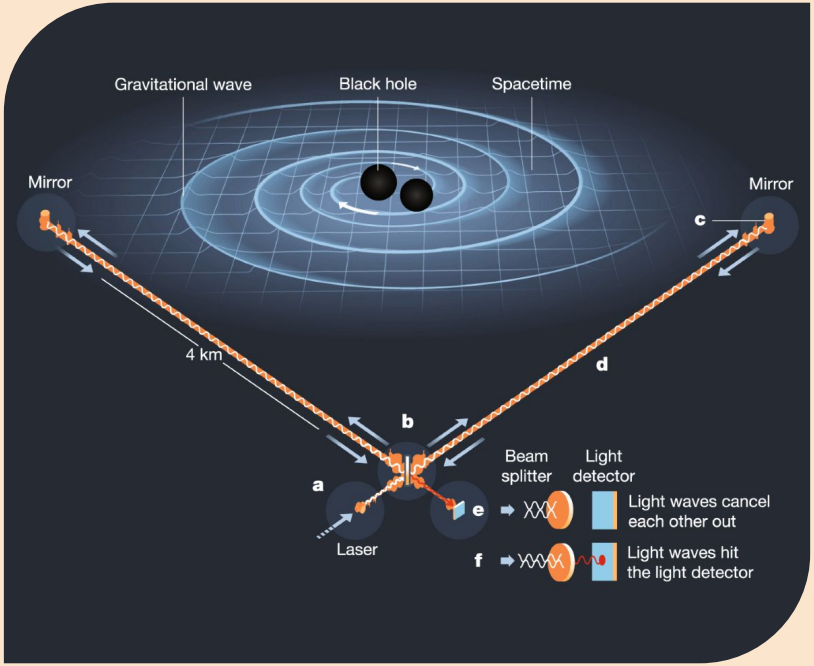
Light curves of various different types of binary systems (not specifically DWD systems)

Kupfer, Thomas, et al. "LISA Galactic Binaries with Astrometry from Gaia DR3." arXiv preprint, arXiv:2302.12719, 2023

Verification Sources for LISA

DWD systems found through electromagnetic (EM) observations act as verification binaries for space based gravitational wave observations like LISA

- Because their sky position, orbital periods and physical properties are already constrained by the optical, their predicted GW signals can be used to validate instrument performance.



Simple diagram of a gravitational wave detector

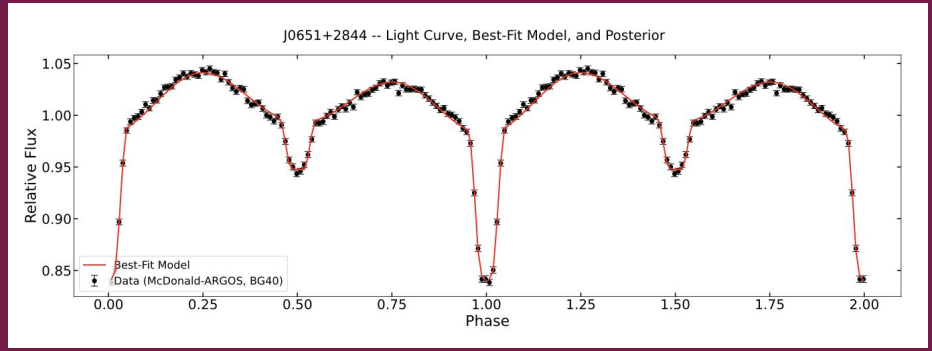
Danzmann, Karsten, et al. *LISA: Laser Interferometer Space Antenna*. arXiv preprint, arXiv:1702.00786, 2017

Characterizing Systems for potential LISA sources

Observing binary systems in the optical typically shows modulation of the systems, this can be eclipses, ellipsoidal various, doppler beaming and reflections.

There are three types of sources:

- **Eclipsing systems:** Geometry directly constraints masses, radii, and inclination to percent-level precision
- **Ellipsoid systems:** Tidal modulations provide partial constraints when paired with spectroscopic priors



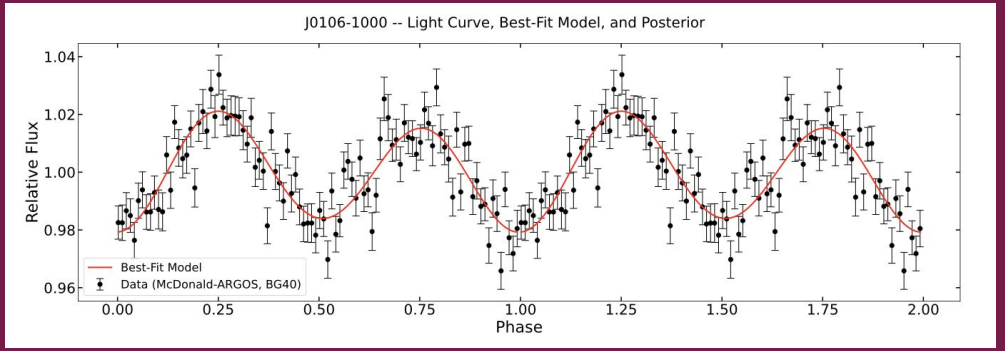


Figure 1: system J0651+2844 light curve of Eclipsing system
Figure 2: system J0106+1000 light curve of Ellipsoid system

Barrientos Laurence, Manuel H. *"Progress Report: Electromagnetic and Gravitational Wave Characterization of LISA Binaries in the Extremely-Low Mass Survey."* University of Oklahoma, 2025. Research Progress Report

Characterizing Systems for potential LISA sources

There are three types of sources:

- **Featureless systems:** Flat optical light curves leave inclination unconstrained and parameters strongly degenerate

It is harder to characterize these systems due to their strong modulations. Unconstrained inclinations lead to massive error bars.

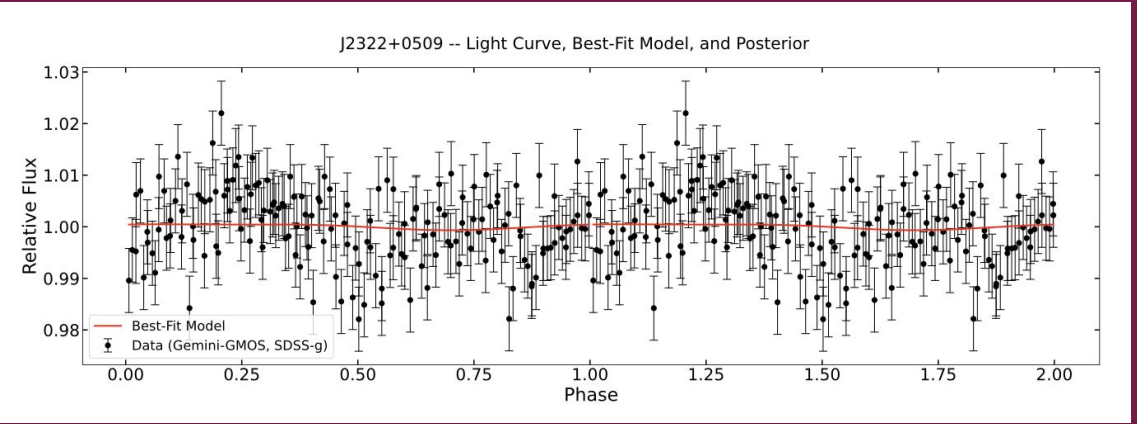


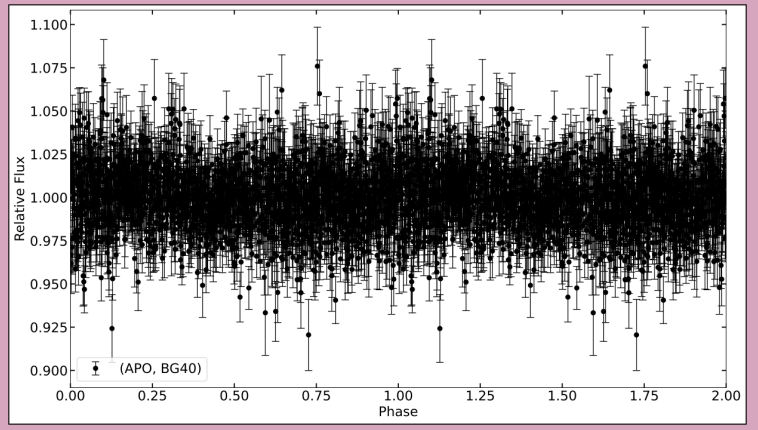
Figure: system J2322+0509 light curve of Featureless system
Red line (straight line) shows the best fit of the system

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How to characterize systems with bad light curves?

The Majority Problem: ~54% of DWD light curves in optical surveys are featureless or poorly modulated

How do we break degeneracies when optical light curves give us almost no geometric modulation?



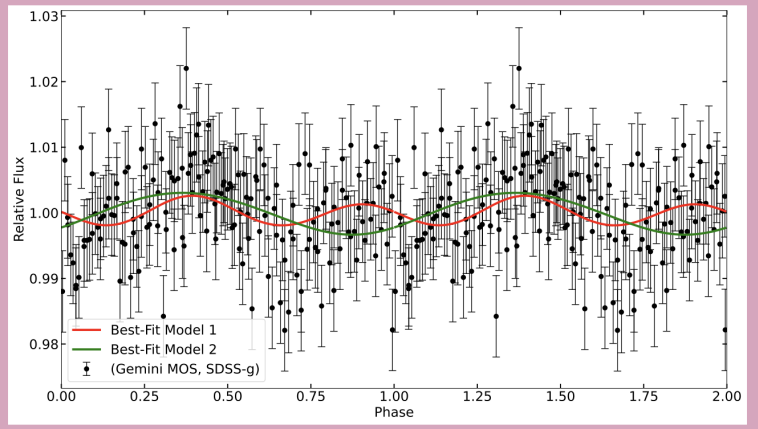


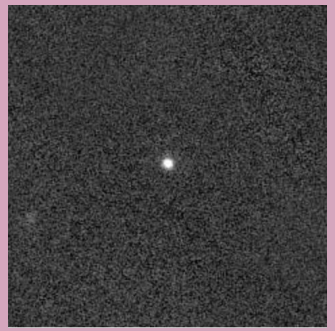
Figure 1: System J0634+3803, 26.55 min period, No clear modulation, Low-inclination system, with degenerate solution.
Figure 2: System J2322+0509, 20.02 min period, No clear modulation, Low-inclination system, with degenerate solution.

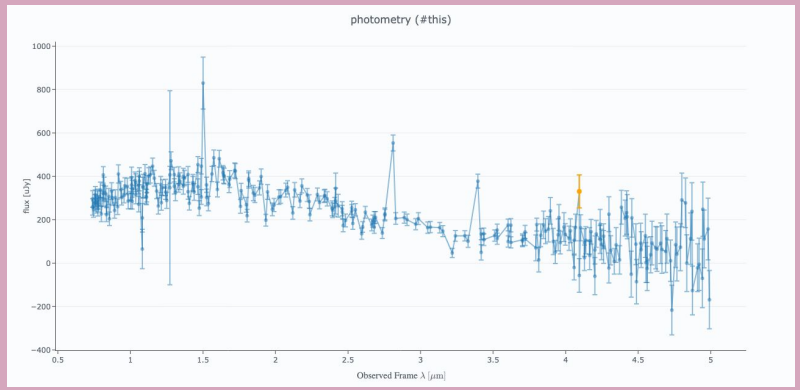
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Inferred observation of the system

Observing at longer wavelengths (e.g., Deep IR ~960 nm) shifts flux ratio balance toward the cooler companion

By observing one of the systems in the inferred we can apply physical parameter constraints to complement the UV and optical constraints of the primary source





System: J0935+4411
Image from PanSTARRS-1 Image Access, in the Deep Infrared, ~960 nm

https://ps1images.stsci.edu/cgi-bin/ps1cutouts?pos=12%3A35%3A49.853+++15%3A43%3A18.954&filter=color&filter=g&filter=r&filter=i&filter=z&filter=y&filetypes=stack&auxiliary=data&size=240&output_size=0&verbose=0&autoscale=99.500000&catlist=

Summary

Main Limitation: ~54% of DWD LISA candidate sources suffer from featureless optical light curves and degenerate parameters.

Infrared Advantage: observations in the IR uncovers the cooler secondary star, allowing for constraints inclination.

LISA Readiness: Tighter EM constraints provide precise GW priors for verification binary sources to help in calibrated LISA

